



Event notifications

Event notifications send messages to your [queue](#) when data in your R2 bucket changes. You can consume these messages with a [consumer Worker](#) or [pull over HTTP](#) from outside of Cloudflare Workers.

Get started with event notifications

Prerequisites

Before getting started, you will need:

- An existing R2 bucket. If you do not already have an existing R2 bucket, refer to [Create buckets](#).

- An existing queue. If you do not already have a queue, refer to [Create a queue](#).
- A [consumer Worker](#) or [HTTP pull](#) enabled on your Queue.

Enable event notifications via Dashboard

1. In the Cloudflare dashboard, go to the **R2 object storage** page.

Go to **Overview** 

2. Select the bucket you'd like to add an event notification rule to.
3. Switch to the **Settings** tab, then scroll down to the **Event notifications** card.
4. Select **Add notification** and choose the queue you'd like to receive notifications and the [type of events](#) that will trigger them.

5. Select **Add notification**.

Enable event notifications via Wrangler

Set up Wrangler

To begin, install [npm ↗](#). Then [install Wrangler, the Developer Platform CLI](#).

Enable event notifications on your R2 bucket

Log in to Wrangler with the [wrangler login command](#). Then add an [event notification rule](#) to your bucket by running the [r2 bucket notification create command](#).

```
npx wrangler r2 bucket notification create  
<BUCKET_NAME> --event-type <EVENT_TYPE> --  
queue <QUEUE_NAME>
```

To add filtering based on prefix or suffix use the `--prefix` or `--suffix` flag, respectively.

Filter using prefix

```
$ npx wrangler r2 bucket notification create  
<BUCKET_NAME> --event-type <EVENT_TYPE> --  
queue <QUEUE_NAME> --prefix "<PREFIX_VALUE>"
```

Filter using suffix

```
$ npx wrangler r2 bucket notification create  
<BUCKET_NAME> --event-type <EVENT_TYPE> --  
queue <QUEUE_NAME> --suffix "<SUFFIX_VALUE>"
```

*# Filter using prefix and suffix. Both the
conditions will be used for filtering*

```
$ npx wrangler r2 bucket notification create  
<BUCKET_NAME> --event-type <EVENT_TYPE> --  
queue <QUEUE_NAME> --prefix "<PREFIX_VALUE>" -  
-suffix "<SUFFIX_VALUE>"
```

For a more complete step-by-step example, refer to the [Log and store upload events in R2 with event](https://developers.cloudflare.com/r2/buckets/event-notifications/)

[notifications](#) example.

Event notification rules

Event notification rules determine the [event types](#) that trigger notifications and optionally enable filtering based on object prefix and suffix. You can have up to 100 event notification rules per R2 bucket.

Event types

Event type	Description	Trigger actions
object-create	Triggered when new objects are created or existing objects are overwritten.	• PutObject • CopyObject • CompleteMultipart

Event type	Description	Trigger actions
object-delete	Triggered when an object is explicitly removed from the bucket.	• DeleteObject • LifecycleDeletion

Message format

Queue consumers receive notifications as [Messages](#).

The following is an example of the body of a message that a consumer Worker will receive:



```
{
  "account":
  "3f4b7e3dcab231cbfdaa90a6a28bd548",
  "action": "CopyObject",
  "bucket": "my-bucket",
  "object": {
    "key": "my-new-object",
    "size": 65536,
    "eTag": "c846ff7a18f28c2e262116d6e8719ef0"
  },
  "eventTime": "2024-05-24T19:36:44.379Z",
  "copySource": {
    "bucket": "my-bucket",
    "object": "my-original-object"
  }
}
```

Properties

Property	Type	Description
account	String	The Cloudflare a that the event is

Property	Type	Description
		with.
action	String	The type of action triggered the event notification. Example actions include: CopyObject, CompleteMultipartUpload, DeleteObject.
bucket	String	The name of the bucket where the event occurred.
object	Object	A nested object containing details about the object involved in the event.
object.key	String	The key (or name) of the object within the bucket.
object.size	Number	The size of the object in bytes. Note: not the size of the metadata.

Property	Type	Description
		for object-deletes.
<code>object.eTag</code>	String	The entity tag (eTag) of the object. Note: no eTag is returned for object-deletes.
<code>eventTime</code>	String	The time when the event that triggered the notification occurred.
<code>copySource</code>	Object	A nested object containing details about the source of a copied object. This property is only present for events triggered by Copy or Copy-Replace.
<code>copySource.bucket</code>	String	The bucket that contains the source object.
<code>copySource.object</code>	String	The name of the source object.

Notes

- Queues [per-queue message throughput](#) is currently 5,000 messages per second. If your workload produces more than 5,000 notifications per second, we recommend splitting notification rules across multiple queues.
- Rules without prefix/suffix apply to all objects in the bucket.
- Overlapping or conflicting rules that could trigger multiple notifications for the same event are not allowed. For example, if you have an object-create (or PutObject action) rule without a prefix and suffix, then adding another object-create (or PutObject action) rule with a prefix like `images/` could trigger more than one notification for a single upload, which is invalid.

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